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IM1012 – Machine Learning

Predicting Chemical Toxicity Using Machine Learning

A Comparative Analysis of Logistic Regression and Random Forest on the Tox21 Dataset

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Git repository : https://github.com/alena-sava/tox21

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Abstract

This project examines the application of machine learning in predicting chemical toxicity through the Tox21 dataset. It specifically targets the SR-ARE endpoint, which relates to biological activities tied to stress responses. The molecular structures were transformed into 2,048-bit Morgan fingerprints, which were utilized as input features for two classification models: Logistic Regression and Random Forest. Their effectiveness was evaluated using several metrics, including ROC-AUC, precision, recall, F1-score, and accuracy. Random Forest achieved a superior ROC-AUC of 0.795 compared to Logistic Regression's 0.764, indicating it was more effective at distinguishing between active and inactive compounds. Conversely, Logistic Regression excelled in recall and F1-score. These results illustrate that the performance of models can depend on the chosen evaluation metric and emphasize the potential of machine learning in computational toxicology.

Introduction: The Issue of Assessing Chemical Safety

The relevance of the research topic

The rapid increase in the number of chemicals used in industry, medicine, and daily life creates a need for effective methods to evaluate their potential toxicity. Traditional approaches, which rely on mathematical regression and experimental studies, have limitations and do not always account for the complex mechanisms of biological interactions with chemical compounds. (Koutsoukas et al. 2016a; Krewski et al. 2010; Tice et al. 2013; Toxicology for the 21st Century—Mapping the Road Ahead | Toxicological Sciences | Oxford Academic n.d.)

Despite the evolution of machine learning techniques and the implementation of neural networks, intricate predictive models that can take into account the links between chemical structures and their biological effects are still not sufficiently advanced. (Cherkasov et al. 2014; Hemmerich and Ecker 2020; Kappenberg et al. 2026; Li et al. 2025).

Tox21 dataset

In this context, the Toxicology in the 21st Century (Tox21) initiative is of special interest as it aims to advance alternative methods for toxicity evaluation. By using high-throughput screening and machine learning, it can analyze a vast number of chemical compounds and predict their possible toxic impacts(Luo et al. 2024; Lynch et al. 2024; Merrick, Paules, and

Tice 2015; Modelling the Tox21 10 K chemical profiles for in vivo toxicity prediction and mechanism characterization | Nature Communications n.d.)..

This dataset offers experimental activity data that can help in creating machine learning models aimed at predicting chemical toxicity and spotting potentially dangerous compounds.

The Tox21 dataset includes around 10,000 chemical compounds, which consist of both environmental chemicals and drugs that have been approved. These compounds were tested using high-throughput screening assays to explore their biological effects on 12 different biochemical pathways: seven related to Nuclear Receptors and five concerning Stress Responses (Koutsoukas et al. 2016b).

Computational toxicology

Computational toxicology merges computational techniques, mathematical modeling, and machine learning to forecast the possible toxic impacts of chemical substances. Traditional methods for testing toxicity, such as in vitro and in vivo experiments, can take a lot of time and money. Machine learning provides a different way by examining large datasets and spotting patterns between chemical structures and their biological effects. Different ML and deep learning techniques, like Random Forest, Support Vector Machines, and Neural Networks, are being used more and more to predict toxicity and biological activity (Guo et al. 2023; Li et al. 2025). These methods allow for the examination of thousands of chemical compounds across various biological tests. Even with notable advancements, there are still challenges, especially concerning data quality, how interpretable the models are, and the reliability of predictions (Guo et al. 2023; Li et al. 2025).

Project aim

The goal of this project is to see if the non-linear Random Forest model offers any benefits compared to a simple baseline model for predicting SR-ARE activity based on molecular fingerprints.

The main goal of this project is to explore if the information found in molecular structures can help predict biological activity in toxicological tests. We represent molecular structures with Morgan fingerprints and use them as input for machine learning models. Two different modeling techniques, Logistic Regression and Random Forest, are being examined. We will then assess their effectiveness in differentiating between compounds with positive and negative SR-ARE activity using suitable classification metrics like ROC-AUC and PR-AUC.

Research Question

Can the non-linear Random Forest model achieve superior performance compared to a Logistic Regression baseline in forecasting SR-ARE activity from molecular fingerprints?

Hypothesis: The Random Forest model is expected to surpass Logistic Regression because it has the ability to recognize non-linear relationships between molecular fingerprint features and SR-ARE activity.

Data Analysis

Dataset

The dataset for this project is the Tox21 toxicological screening dataset, which consists of 7,831 chemical compounds and 12 biological activity endpoints, including nuclear receptor (NR) and stress response (SR) assays. Each compound is represented by a SMILES string that gives a textual description of its molecular structure. The dataset also includes a compound identifier (mol_id) and experimental activity measurements for the various endpoints(Chen et al. 2021a; Wu et al. 2018).

In this project, SR-ARE was chosen as the target endpoint. The prediction task is framed as a binary classification, where 0 signifies a negative experimental result and 1 signifies a positive result. The original SR-ARE measurements included 4,890 negative observations, 942 positive observations, and 1,999 missing values (Chen et al. 2021b). Missing SR-ARE values were not considered negative observations because the lack of a measurement does not indicate biological inactivity.

Data preprocessing and cleaning

Before building the machine learning features, data preprocessing took place. The SR-ARE and other endpoint columns were changed into numeric values. RDKit was used to validate molecular structures by transforming the SMILES strings into molecular objects. Any invalid SMILES representations were detected and eliminated.

Canonical SMILES and InChIKeys were created to give standardized views of the molecular structures and help in finding duplicates. Additionally, we checked duplicate molecular structures for any conflicting SR-ARE labels. There were no conflicting SR-ARE labels found in the duplicate groups we looked at. After picking compounds that had an SR-ARE measurement available, we ended up with 5,633 labeled compounds. Out of these, 4,728 were negative and 905 were positive. This means that about 16% of the labeled observations were in the positive class, showing a class imbalance. We took this imbalance into account during model development by applying class-balanced weighting for both Logistic Regression and Random Forest.

Molecular representation

The confirmed molecular structures were converted into Morgan fingerprints with a fingerprint size of 2,048 bits. This meant that each compound was represented by a fixed-length binary feature vector. The resulting feature matrix had 5,633 compounds and 2,048 molecular features.

Next, the data was split into training and test sets with an 80/20 ratio. Stratification was used to maintain the balance of positive and negative SR-ARE observations in both groups.

Methodology and Implementation

This project uses a comparative machine learning method. The primary goal is to see if a nonlinear model can better distinguish SR-ARE activity compared to a linear baseline, given that both models utilize the same molecular representation.

Logistic Regression was chosen as the baseline model due to its straightforward linear classification method. Random Forest was picked as the second model because it can capture nonlinear relationships between molecular fingerprint features and the target variable. This setup enables an exploration of how model complexity affects results while maintaining a consistent input representation.

To make sure the comparison is fair, both models were trained on the same Morgan fingerprint features and tested on the same held-out test set. The labeled dataset was split into training and test subsets with an 80/20 ratio, using stratification. Stratification was necessary because the SR-ARE target has an imbalance, with many more negative observations than positive ones.

Morgan fingerprints bits were used as the molecular representation. Each fingerprint captures the structural patterns of the corresponding molecule as a fixed-length binary feature vector, which allows the chemical structures represented by SMILES to be input into standard machine learning algorithms.

Model performance was measured using ROC-AUC, precision, recall, F1-score, and accuracy. ROC-AUC was chosen as the main metric for discrimination because it assesses how well positive and negative observations are ranked at various classification thresholds. Precision, recall, and F1-score were also important to consider since the positive SR-ARE class is relatively rare. The models also produced class probabilities that were used for the ROC-AUC evaluation.

Results

Logistic Regression reached a ROC-AUC of 0.764, whereas Random Forest scored a higher ROC-AUC of 0.795. This suggests that Random Forest was more effective at distinguishing between compounds with positive and negative SR-ARE activity.

Additionally, Random Forest had better precision (0.617 compared to 0.398) and accuracy (0.849 versus 0.795). On the other hand, Logistic Regression demonstrated significantly higher recall (0.536 against 0.160) and a superior F1-score (0.456 compared to 0.254).

The two models demonstrated varying performance traits. Random Forest excelled in accurately identifying correct positive predictions, while Logistic Regression managed to recognize a greater share of the real positive compounds at the standard classification threshold of 0.5.

Model	ROC-AUC	Precision	Recall	F1	Accuracy
Logistic Regression	0.764025	0.397541	0.535912	0.456471	0.795031
Random Forest	0.795493	0.617021	0.160221	0.254386	0.849157

Conclusion

This study examined if a nonlinear Random Forest model could outperform a Logistic Regression baseline in distinguishing SR-ARE activity, both using the same Morgan fingerprint representation.

The Random Forest model achieved a higher ROC-AUC score than the Logistic Regression model (0.795 vs. 0.764), suggesting it was better at telling apart active and inactive compounds. On the other hand, Logistic Regression had much higher recall and F1-score at the default threshold of 0.5.

Therefore, the results lend some support to the hypothesis that a nonlinear model can enhance prediction performance. Random Forest showed superiority in overall discrimination, but Logistic Regression was more successful in detecting positive SR-ARE compounds at the specified classification threshold.

Discussion

The increased ROC-AUC from Random Forest indicates that this nonlinear model was more effective at differentiating between compounds with and without SR-ARE activity when looking at predictions across various classification thresholds.

This suggests that the connections between molecular substructures shown by the Morgan fingerprints and SR-ARE activity are likely not just linear. Random Forest is capable of modeling nonlinear relationships and interactions among fingerprint features, while Logistic Regression only offers a linear decision function.

However, despite Random Forest's higher ROC-AUC, it did not lead to improved recall or F1-score at the default threshold of 0.5. Logistic Regression performed significantly better in terms of recall and F1-score. This shows that the performance of a model is contingent on the evaluation metric and classification threshold, and that ROC-AUC alone does not encompass all facets of classification performance.

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